ANURAG UNIVERSITY | SCHOOL OF ENGINEERING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AI-Powered Campus Management System

Abstract:

The AI-Powered Campus Management System is a comprehensive web-based platform designed to digitally integrate and simplify essential academic and administrative activities within a college environment. The system provides separate role-based dashboards for students, faculty, HODs, placement officers, and administrators, enabling secure access to personalized features such as student profiles, subjects, attendance, marks, assignments, examinations, notices, events, grievances, and placement activities. The platform is developed using React and Vite for the frontend and Java Spring Boot for the backend, with REST APIs, MySQL, JPA, JWT authentication, Spring Security, and role-based access control ensuring secure and efficient data management. An AI-powered college assistant is integrated using Large Language Models, embeddings, vector search, and Retrieval-Augmented Generation (RAG) to provide context-aware answers based on college-specific information and documents, along with conversation history and personalized assistance. The system further incorporates caching, asynchronous processing, logging, monitoring, testing, Docker-based containerization, API documentation, and cloud deployment to improve performance, reliability, scalability, and maintainability. Overall, the proposed system provides a centralized, intelligent, secure, and scalable digital campus environment that enhances communication, academic management, administrative efficiency, and student support.

